

Title Estimating duration in sex work from cross-sectional studies

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Abstract Average total durations selling or buying sex are key determinants of HIV/STI transmission dynamics and service coverage. Such durations are usually estimated from cross-sectional surveys of people actively selling/buying sex. However, cross-sectional surveys tend to oversample individuals selling/buying sex for longer, while survey-reported durations will be right-censored. Prior works have not precisely addressed these two competing biases. In this paper, we derive the exact relationship between the distribution of durations reported in cross-sectional surveys and the distribution of total durations in the population overall. These distributions will be identical if total durations are exponentially distributed; otherwise, reported durations will typically underestimate the variance in total durations. We then develop a parametric Bayesian model of this relationship, and apply it to re-analyze aggregate data from a global systematic review. Our model-estimated average total durations are significantly shorter than previously published for all regions — e.g. in Africa: 3.7 (95% CI: 3.4–3.9) versus 5.5 years; in Asia: 1.4 (1.2–1.5) versus 2.8–4.0 years. Shorter average durations then suggest that HIV incidence in sex work could be higher than previously thought, to match the same HIV prevalence data. Our approach can be applied beyond sex work, to characterize duration distributions for any transient risk/disease state.

Keywords censoring, length-time bias, HIV/AIDS, STI, sex work

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1 Introduction

Around the world, people engaged in sex work — selling or buying sex for money or equivalent — experience high risks of HIV acquisition and transmission due to intersecting structural, behavioural, and biological risk factors [1–5]. Unmet HIV prevention and treatment needs among these populations have long been recognized as drivers of epidemic emergence and persistence, especially through insights from mathematical models of HIV transmission [6–8]. Yet, engagement in sex work is often transient, with entry (for sellers) driven primarily by economic and social instability, and exit further mediated by aging, stable partnering, stigma, and violence [9–11].

As such, durations selling or buying sex are key determinants of HIV epidemiology, and key inputs to HIV transmission models, particularly those aiming to compare the impacts of different prioritized services. For example, 10% HIV prevalence among sex workers in a context with shorter typical durations / rapid turnover (e.g. 1 year) suggests a high HIV incidence rate, whereas 10% HIV prevalence in a context with longer typical durations / slower turnover (e.g. 10 years) suggests a relatively lower HIV incidence rate, since prevalence has more time to accumulate [12].¹ Turnover can also moderate levels of program / service coverage among sex workers (e.g. PrEP), where rapid turnover could result in a constant influx of new sex workers to reach, and potentially high loss-to-follow-up among those exiting sex work,² whereas slower turnover could make it easier to attain higher levels of coverage. Finally, duration in sex work can further moderate onward transmission risk, where longer durations theoretically allow more onward transmission following acquisition (in the absence of treatment).

Ideally, context-specific estimates of sex work duration could be obtained from representative longitudinal cohorts with minimal loss-to-follow-up, or from representative samples of "retired" sex workers.³ Unfortunately, such study designs are rare, and most estimates derive from cross-sectional surveys of those actively selling or buying sex [13]. Yet, it is not necessarily obvious how durations reported in cross-sectional surveys relate to durations of interest, as we will discuss in detail herein. As one recent HIV modelling study noted when specifying plausible parameter bounds: *"We multiply the lower bound by 0.5 and upper bound by 2 because of uncertainty in how overall duration of sex work relates to current duration of sex work"* [14].

Objectives To address this gap, we sought to offer guidance on estimating duration in sex work from commonly available data. Specifically, our objectives were to:

1. Describe the exact analytic relationship between the distribution of reported durations selling sex in cross-sectional studies, and the distribution of total durations selling sex in the population overall.
2. Develop and apply a Bayesian model which leverages the above relationships to infer the distribution of total durations selling sex in the population overall from published data.

¹ Assuming HIV prevalence among new sex workers is negligible.

² Questions remain regarding to what degree those exiting sex work remain engaged with programs.

³ Throughout: we often refer to "sex workers" for brevity, but our statements usually apply similarly to their clients who buy sex; we intentionally use gender-agnostic terminology, since men, women, and transgender individuals can all sell and buy sex.

While we focus on sex work as an illustrative example, our insights likely extend to other transient risk/disease states relevant to compartmental modelling or epidemiological study.

2 Preliminaries

Assumptions For simplicity, we first assume that periods of sex work can be precisely and instantaneously delineated. That is, we assume clear boundaries between formal sex work and transactional sex, and that surveys predominantly reach those actively engaged in sex work, despite some surveys including anyone who bought/sold sex anytime in the past year, etc. [1]. We further assume that survey participants can accurately report how long they have bought/sold sex — i.e. perfect recall, no social desirability bias. Finally, we assume that all populations are at equilibrium, e.g. rates of entering/exiting sex work are stable over time.

Existing estimation methods In their global scoping review of HIV risk behaviour durations, Fazito et al. [13] considered three methods to estimate mean duration selling/buying sex (Table 1): 1. mean total duration among those finished selling/buying sex; 2. reciprocal of the proportion who started selling/buying sex within the past year; and 3. double the mean/median reported duration among those actively selling/buying sex. Notably, among the 118 estimates of duration selling or buying sex identified by Fazito et al.: the vast majority used method 3 (86%), a small fraction used method 2 (14%), and none used method 1. Studies also commonly report the proportions of respondents selling/buying sex for pre-specified duration ranges, e.g. {0–1, 1–5, 5+} years, though it's not clear if/how such data were used in [13].

Who exactly are we talking about? Before continuing, it is helpful to distinguish two populations: individuals who will ever sell/buy sex in their lifetime, and individuals who are actively selling/buying sex. The former "source population" is the sampling frame for method 1, and tends to include more individuals who sell/buy sex for shorter durations; the latter "active population" is the sampling frame for methods 2 and 3, and tends to include more individuals who sell/buy sex for longer durations (Figure 1). Since "sex work" strata in compartmental models reflect the active population, we might assume that rates of exit from sex work should be derived from the active population. However, this is not the case, because these exit rates apply to all individuals *entering* sex work, which reflects the source population.⁴ A final population worth distinguishing is the "sampled population" in a given cross-sectional survey, which we assume intends to be representative of the active population, but might undersample some individuals, e.g. those in the early years of sex work, due to delays in self-identification as a sex worker [15, 16].

⁴ An analogous example: the mean duration of an infection should be characterized by a cohort of individuals with *incident* infections, not a cross-section of individuals with *prevalent* infections.

Table 1: Methods of estimating mean duration selling sex suggested and reviewed in [13]

	Definition	Potential challenges	N [†]
1.	mean total duration among those finished selling sex	• very difficult to sample this population	0
2.	reciprocal of the proportion who started selling sex within the past year among those actively selling sex	• this proportion is rarely reported • sensitive to delays in self-identification • not adjusted for sampling bias	16
3.	double the mean/median reported (right-censored) duration among those actively selling sex	• median < mean for skewed distributions • not adjusted for sampling bias	93

[†] Number of estimates of duration selling sex synthesized by [13]; corresponding numbers were: 0, 1, 8 for duration buying sex.

3 Analytic relationships

We now turn to deriving the exact relationship between: the distribution of *total* durations selling sex among the source population: $x \sim p(x)$, and the distribution of *observed* durations selling sex among participants in a cross-sectional survey: $z \sim p(z|s)$, given survey participation status s . We also consider two survey sampling cases: an ideal case which is representative of the active population, and a more plausible case which undersamples individuals in the early years of sex work. For reference throughout this section, Figure 2 illustrates example distributions for $p(x)$, $p(z|s)$, and intermediates thereof.

Length-biased sampling The probability of an individual being active at any given time is proportional to their total duration x ; a phenomenon well-described in cancer epidemiology as "length-biased sampling" [17]. Thus, the distribution of total durations among the active population is [18]:

$$p(x|a) = \frac{1}{K} x p(x) \quad (1)$$

where K is a normalizing constant. These total durations among the active population are typically unobserved, except in cohort studies which follow participants until they exit sex work. From Eq. (1) we can see that the mean total duration among the active population $\mathbb{E}[x|a]$ is always larger than the mean among the source population $\mathbb{E}[x]$ (Figure A.5, light green vs gray circles); specifically, we have that [18]:

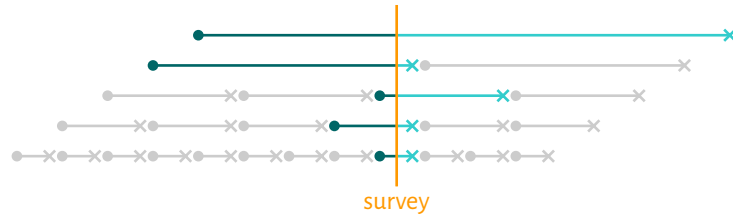


Figure 1: Diagram of periods selling/buying sex in an open population at equilibrium

Colours — dark green: past sex work among active population (observed); light green: future sex work among active population (unobserved); gray: past/future total sex work periods among inactive population (unobserved).

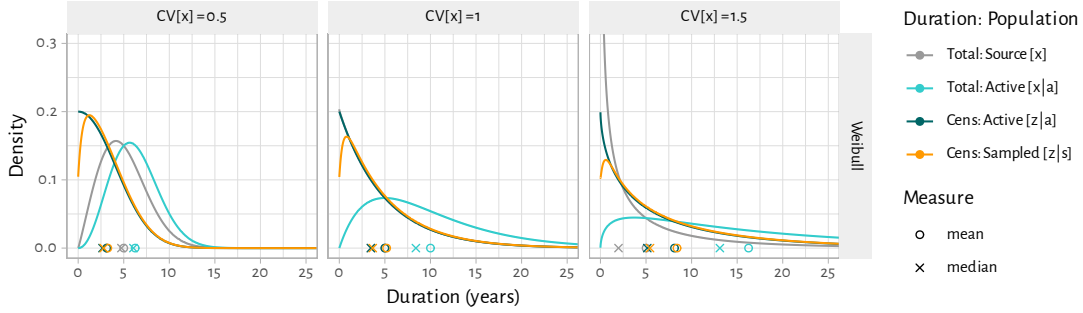


Figure 2: Illustrative distributions of total and censored durations among the source, active, and sampled populations for 3 coefficients of variation $CV[x]$ (rows)

Assumed: $E[x] = 5$ and $x \sim \text{Weibull}$ (Figure A.5 illustrates other distribution families); probability of being sampled per Eq. (7).

$$E[x|a] = E[x] (1 + CV^2[x]) \quad (2)$$

This is because greater variability in durations among the source population, as measured by the coefficient of variation: $CV[x]$, allows those with longer durations to increasingly dominate the active population. We also note that the probability of observing active individuals with very short total durations is inherently very small (Figure 2, light green).

Censoring If survey timing is effectively random, then observed durations at the time of survey (z) will be right-censored and uniformly distributed between 0 and the total duration (x):

$$p(z|x) = \text{Unif}(0, x) = \begin{cases} \frac{1}{x} & \text{if } 0 \leq z \leq x \\ 0 & \text{else} \end{cases} \quad (3)$$

Since the mean of $\text{Unif}(0, x)$ is $\frac{x}{2}$, the mean censored duration is always half the mean total duration among the active population: $E[z|a] = \frac{1}{2}E[x|a]$ (Figure 2, dark green vs light green circles). This idea appears to be well-known, and likely motivates doubling the mean $E[z|a]$ to estimate $E[x]$ in method 3 of [13] — i.e. to "undo" halving the mean. However, method 3 does not account for selection of the active population noted above, and thus actually estimates the mean total duration among the active population $E[x|a]$, which we know to be greater than among the source population $E[x]$ per Eq. (2).⁵

If total durations among the source population are exponentially distributed: $p(x) \sim \text{Exp}(\lambda)$, then $CV[x] = 1$, and so the mean *censored* duration among the *active* population will directly equal the mean *total* duration among the *source* population (without doubling): $E[z|a] = E[x]$. In other words, under these conditions, the mean reported duration in a typical cross-sectional survey will be an unbiased estimator of the target $E[x]$. Moreover, even the *distributions* will be identical: $p(z|a) = p(x) = \text{Exp}(\lambda)$, as we will show below. This is

⁵ Fazito et al. [13] partially acknowledge this issue by opting to use the median over the mean of $z|a$ where possible, since the median is less sensitive to outliers. However, as shown in Figures A.5 and A.6, doubling the median $M[z|a]$ still overestimates $E[x]$, just to a lesser degree than doubling the mean $E[z|a]$, since the median will always be less than the mean for right-skewed distributions.

particularly notable since compartmental models usually assume exponentially distributed durations.

Complete distribution By combining Eqs. (1) and (3), we can derive the complete distribution of censored durations among the active population $p(z|a)$ from the distribution of total durations among the source population $p(x)$ [19]:

$$\begin{aligned}
 p(z|a) &= \int_0^\infty p(z|x) p(x|a) dx \\
 &= \int_z^\infty \frac{1}{x} p(x|a) dx \\
 &= \frac{1}{K} \int_z^\infty \frac{x}{x} p(x) dx \\
 &= \frac{1}{K} (1 - F_x(z))
 \end{aligned} \tag{4}$$

As it turns out, this is simply one minus the cumulative distribution of total durations F_x , normalized to have unit area under the curve (via K).⁶ For the exponential case, we have: $F_x = 1 - e^{-\lambda x}$, $K = \lambda^{-1}$, and thus $p(z|a) = \lambda e^{-\lambda z} = p(x)$.

More generally, $\text{CV}[z|a]$ is typically drawn towards 1 relative to $\text{CV}[x]$, attenuating skewness when $\text{CV}[x] > 1$, and amplifying skewness when $\text{CV}[x] < 1$ (Figures 2 and A.5). However, we believe that $\text{CV}[x] > 1$ is more likely than $\text{CV}[x] < 1$ for two reasons. First, heavy-tailed distributions having $\text{CV}[x] > 1$ arise naturally via selection effects in populations with heterogeneous rates [20]. Second, most parametric distributions with $\text{CV}[x] < 1$ have $p(x \rightarrow 0) \rightarrow 0$ (Figure A.5), which effectively implies that few people exit sex work shortly after starting, or that most people stay in sex work for a "minimum" period, which seems unrealistic. If $\text{CV}[x] > 1$ is indeed more likely, then the skewness of the observable duration distribution $p(z|a)$ will generally underestimate the skewness of the true duration distribution $p(x)$.

Further sampling bias If the survey further undersamples individuals in the early years of sex work — i.e. beyond effects of length-based sampling, due to delays in self-identification as a sex worker — this probability of being sampled will be a function of current duration (z) not total duration (x), so we denote it $p(s|z)$. Many parameterizations are possible; e.g. Eq. (7). The distribution of observed durations among survey participants is then updated from Eq. (4) simply as:

$$p(z|s) = \frac{1}{K} p(s|z) (1 - F_x(z)) \tag{5}$$

Estimating $\mathbb{E}[x]$ We conclude this section by revisiting the three methods considered by Fazito et al. [13] to estimate the "true" mean duration in sex work $\mathbb{E}[x]$. Figure A.6 plots: the typically unobserved mean and median of total durations (method 1), *double* the means and medians of censored durations (method 3), and the reciprocal proportion of censored durations less than 1 year (method 2), across a range of $\text{CV}[x]$ and distribution families. We see that at low $\text{CV}[x]$, nearly all methods converge on $\mathbb{E}[x]$. However, as $\text{CV}[x]$ increases, selection towards longer total durations among the active or sampled populations raises the means of the corresponding censored durations $\mathbb{E}[z|\cdot]$, according to Eq. (2). Median censored durations exhibit a

⁶ The integral lower bound changes from 0 to z because $p(z|x) = \frac{1}{x}$ only when $x < z$ and $p(z|x) = 0$ otherwise, per Eq. (3).

similar trend, but lag below means (true for all right-skewed distributions) to a degree which depends on the family, making such medians less useful estimators of $\mathbb{E}[x]$. For all method 3 estimates, undersampling of early sex work (here assuming $\rho_s = 0.5$, $\tau_{95} = 1$ year) predictably biases estimates slightly higher; this bias is also larger for smaller $\mathbb{E}[x]$ (not shown). Finally, turnover-based estimates (method 2) tends to be most consistent, only slightly overestimating $\mathbb{E}[x]$ as $\text{CV}[x]$ increases; however, this method is especially sensitive to undersampling of early sex work, which biases estimates upwards.

Nevertheless, the best simple estimate of $\mathbb{E}[x]$ is likely $\mathbb{E}[z|s]$, assuming $\text{CV}[x] \approx 1$, or more generally:

$$\hat{\mathbb{E}}[x] = \frac{1}{2} \mathbb{E}[z|s] (1 + \text{CV}^2[x]) \quad (6)$$

4 Application

We demonstrate the utility of the above analytic relationships through a Bayesian model, which aims to infer the complete distribution of total durations among the source population $p(x)$ from individual-level and/or aggregate data on the distribution of censored durations among the sampled population $p(z|s)$. Specifically, we focus on the types of aggregate estimates reviewed by Fazito et al. [13]: distribution means $\hat{\mathbb{E}}[z|s]$, and proportions of durations within a given interval $[a, b]$, denoted $\hat{\mathbb{P}}[a < z < b|s]$, which can encompass proportions of responses falling within pre-specified ranges, as well as estimated medians, quartiles, etc.

4.1 Bayesian Model

The model assumes a parametric distribution for total durations among the source population $p(x)$. Specifically, we considered 6 distribution families: Exponential, Gamma, Weibull, Log-Normal, Scaled Beta, and Scaled Kumaraswamy, all truncated or scaled to 50 years maximum duration (Table A.1). For convenience, we denote the parameters of all families as: α, β (for Exponential: β only), so the interpretation of and priors for each parameter depend on the family.

We model the probability of being sampled for survey participation among the active population as increasing exponentially from $\rho_s \in [0.5, 1]$ initially to 95% by year $\tau_{95} \in [0, 3]$, and 100% asymptotically:⁷

$$p(s|z) = 1 + (\rho_s - 1) e^{\left(\frac{z}{\tau_{95}} \log \left(\frac{0.05}{1 - \rho_s} \right) \right)} \quad (7)$$

We then define the distribution of censored durations among the sampled population $p(z|s; \alpha, \beta, \rho_s, \tau_{95})$ via Eq. (5). What remains is to specify likelihood functions for estimates $\hat{\mathbb{E}}[z|s]$ and $\hat{\mathbb{P}}[a < z < b|s]$ from empirical data, which are defined in Eqs. (A.1) and (A.2), respectively.

Under this model, we infer posterior distributions for $\alpha, \beta, \rho_s, \tau_{95}$ given the data. From α, β , we can then

⁷ Thus, if either $\rho_s = 1$ or $\tau_{95} = 0$, then sampling will be modelled as perfectly representative of the active population: $p(s|z) = 1$.

compute the distribution of total durations among the source population $p(x; \alpha, \beta)$, and any characteristics thereof, e.g. the mean $\mathbb{E}[x; \alpha, \beta]$. We implemented and fit the model using Stan [21], retaining 1000 samples after 500 warm-up iterations from 7 chains with default random initialization and sampling algorithm [22]. We confirmed convergence visually through parameter trace plots (not shown).

4.2 Aggregate data from a single survey

We first apply the model to aggregate data from a single survey: a 2011 cross-sectional survey of 328 female sex workers in Eswatini [23]. The authors reported crude and respondent-driven sampling (RDS)-adjusted mean (95% CI) proportions of respondents who had been selling sex for 0–3, 3–6, 6–11, and 11+ years (Table A.2). We converted these 95% CI to effective sample sizes per stratum (N_i), for compatibility with Eq. (A.2).

Figure 3 illustrates (i) the posterior distributions of modelled proportions vs the RDS-adjusted data, and (ii) the corresponding estimated distribution $p(z|s)$ for the best-fitting family (Gamma). Figure A.7 further illustrates the posterior distributions of model parameters and summary statistics for $p(x)$ and $p(z|s)$ across all families considered. We find that the model can reproduce these data quite well. However, considerable uncertainty remains regarding the proportions of respondents selling sex for less than 1 year because the available data have aggregated the first 0–3 years into one stratum. This aggregation issue also limits our ability to infer parameters of the sampling model: ρ_s, τ_{95} , whose posterior distributions are effectively indistinguishable from their priors.

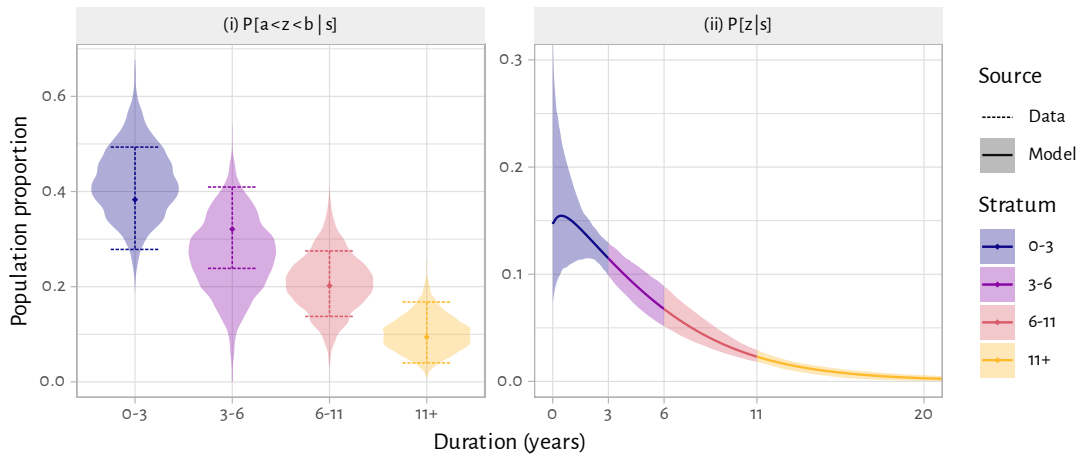


Figure 3: Posterior estimates from fitting the Bayesian model to RDS-adjusted aggregate censored duration data from [23]: (i) proportions of respondents with durations in pre-specified ranges; (ii) inferred duration density distribution $p(z|s)$

Shaded regions denote: (i) posterior of modelled proportions; (ii) posterior 95% CI of modelled duration distribution. Points and dashed line ranges denote reported 95% CI for proportion data.

The posterior distribution of $\mathbb{E}[z|s]$ was reasonably precise and consistent across the families considered: approximately 5.0 (95% CI: 4.4, 5.8), and likewise for $\mathbb{CV}[z|s]$: 0.9 (0.7, 1.2). By contrast, estimates of $\mathbb{E}[x]$ and $\mathbb{CV}[x]$ were more varied, with 95% CI extremes spanning (2.0, 8.8) and (0.4, 1.9), respectively, and best-fitting $\mathbb{E}[x] = 5.7$, $\mathbb{CV}[x] = 0.9$ under the Gamma family.

4.3 Aggregate data from a global review

Noting the limitations of previously considered methods to estimate $\mathbb{E}[x]$, we additionally applied the model to re-synthesize duration data reviewed by Fazito et al. [13]. We manually extracted aggregate data for female sex workers from Table S2 in [13] on duration means $\mathbb{E}[z|s]$ and interval proportions $\hat{P}[a < z < b|s]$, including survey region, sample size, and other meta-data; see § A.3.2 for additional details. We then ran the model separately for each region with data, and for each distribution family.

Figures A.8–A.11 illustrate the posterior distributions of model parameters, summary statistics, cumulative/density duration distributions, and means versus the data. While censored duration distributions $p(z|s)$ were approximately consistent across families, the inferred total duration distributions $p(x)$ and corresponding means $\mathbb{E}[x]$ depended strongly on the family. In particular, $\mathbb{E}[x]$ was typically longest under Exponential and Log-Normal families and shortest under the Scaled Kumaraswamy family, with $\mathbb{E}[x]$ and $\mathbb{CV}[x]$ negatively correlated. Gamma, Weibull, Scaled Beta, and Scaled Kumaraswamy families also often suggested that a large proportion of women in the source population sold sex for under 1 year. According to total log-likelihoods, the best-fitting families were: Log-Normal for Africa, Weibull for Asia and Europe, and Scaled Kumaraswamy for Latin America; the Exponential family consistently fit worst by failing to capture the "heavy tails" of $p(z|s)$ and $p(x)$. Nevertheless, Exponential modelling of durations likely remains reasonable.

Finally, Figure 4 illustrates the mean (95% CI) for our estimates of mean total duration $\mathbb{E}[x]$ versus those in Table 1 of [13]. Across all distribution families considered, our estimates are consistently shorter than those of [13], with best-fitting families ranging from 1.5x shorter in Africa to 4x shorter in Latin America.

5 Discussion

We sought to clarify several concepts relevant to characterizing the distribution of durations selling or buying sex. First, we distinguished between "source" and "active" populations, with the active population being a length-biased sample of the source population. Second, we reiterated the familiar implications of randomly censored durations, which halves the mean duration. We combined the above to derive the exact relationship between the distribution of censored durations among the active population and the distribution of total durations among the source population, which happen to be exactly equal if duration are Exponentially distributed. We then developed a parametric Bayesian model of this relationship to infer the distribution of total durations among the source population from aggregate data on censored durations among the active population. Applying this model to data from a previous global systematic review, we inferred significantly shorter mean durations across multiple regions.

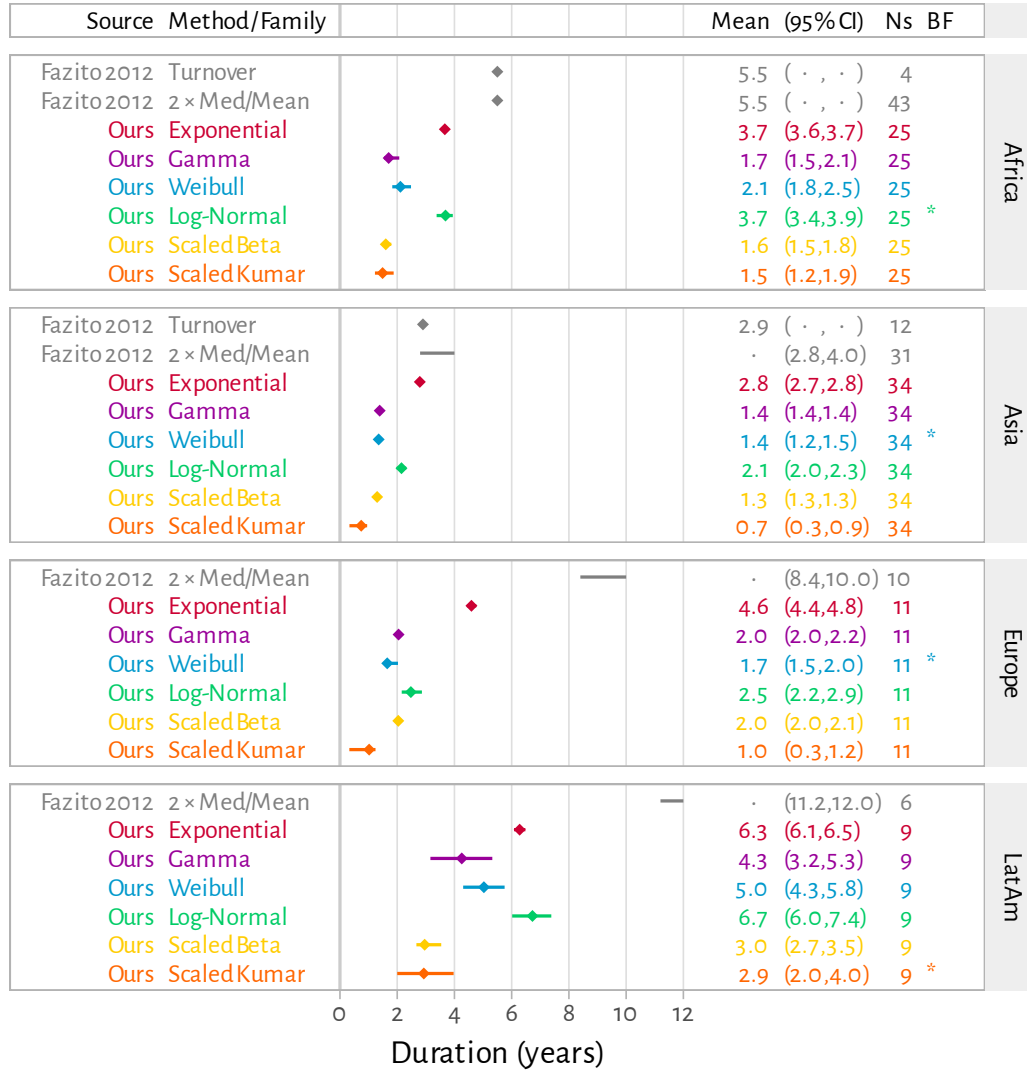


Figure 4: Forest plot of estimated mean total duration among female sex workers $\mathbb{E}[x]$ versus regional synthesis estimates in [13]

Synthesis methodology in [13] was unclear, and regional estimates were variably reported as points or ranges; see Table 1 for method definitions. Ns: number of unique surveys contributing to the estimate; BF: best-fitting family per region.

Our estimates of mean duration selling sex were around half or less versus those reported in [13]. This was expected, as we showed how both methods 2 and 3 can overestimate mean duration, especially if surveys undersample individuals in the early years of sex work. Shorter durations suggest that HIV incidence in sex work could be higher than previously estimated via transmission models, since there is less time for infections to accumulate to match observed HIV prevalence in sex work [12]. However, existing model-based estimates of HIV incidence among female sex workers in Sub-Saharan Africa already exceed empirical estimates [24]; this might be partly explained by recruitment of relatively lower-risk female sex workers and relatively higher-risk wider population women for empirical HIV incidence studies [25]. Rapid sex work turnover could also suggest that more infections in the wider population might be attributable to unmet prevention needs during brief periods of sex work [12].

We also inferred that distributions of total durations in sex work are likely "heavy-tailed", such that rates of exit decline with duration in sex work. This declining exit rate could reflect selection effects acting on between-person differences in propensity to remain in sex work, i.e. "frailties" [20], and/or within-person changes while in sex work, including self-identification as a sex worker, and changing social ties with family, friends, and clients [9–11, 26, 27]. While individual-based simulation models could specify "heavy-tailed" distributions directly, compartmental simulation models would need two or more sex work states to reproduce this pattern — e.g. separate short and long duration sex work trajectories, and/or separate early and late sex work periods. Indeed, such states would also be useful to capture unique risks associated with early sex work [26, 27].

Limitations Our work has several limitations. First, we did not model any gaps (temporary breaks) in sex work. However, many estimates of sex work duration are derived from the difference between respondents' current age and the age they report first selling sex, which implicitly ignores such gaps. Respondents may also report "duration selling sex" in this way when asked directly, especially since unsuccessful attempts to exit sex work can be stigmatized [11]. Second, we did not consider any recall/reporting biases, as entry into sex work is likely a memorable life-course event [9–11, 28, 29], and self-identified sex workers do not, to our knowledge, have obvious motivations for under-reporting durations. We additionally assumed that sex work turnover is stable over time, despite empirical evidence of changing sex work dynamics in several contexts [30, 31]. Finally, our re-analysis of duration data from [13] did not include more recent or individual-level data, since our motivation was mainly to compare results using our proposed versus prior estimation methods. Additionally, in forthcoming work we will synthesize individual-level data from 183 studies [31].

Future work Our Bayesian model could be extended in several ways. First, it might be more useful to model sex work durations via semi/parametric hazard rates of exiting sex work, rather than parametric duration distributions.⁸ A hazard rate approach could allow and incorporation of more intuitive parameter priors (e.g. 0.5–50% exit rate per month), covariates (e.g. under proportional hazards assumptions) [32], and frailties (e.g. to capture unmodelled heterogeneity) [20]. Relatedly, joint modelling of duration and age and/or calendar time could help characterize relevant trends noted above e.g. for HIV/STI transmission modelling [30, 31]. Such extensions might be supported by related work estimating durations (also called sojourn times) in chronic disease states for modelling and analysis of screening programs [18].

⁸ Indeed, the hazard rates implied by some parametric distributions we considered may be implausible, e.g. Log-Normal.

Conclusion «TODO»

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Appendix

Title Estimating duration in sex work from cross-sectional studies

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Appendix A

Supplementary Content

A.1 Parametric equations

Distribution families We considered 6 distribution families for $p(x)$ in our analysis (Table A.1).

Table A.1: Distribution families considered, using common parameter notation (α, β)

Family	Definition $p(x)$	α log-prior [*]	β log-prior [*]
Exponential [†]	$\frac{1}{\alpha} e^{-\frac{x}{\alpha}}$	–	[–1, 3]
Gamma [†]	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$	[–2, 2]	[–3, 3]
Weibull [†]	$\frac{\alpha}{\beta} \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-(x/\beta)^\alpha}$	[–1, 2]	[–1, 2]
Log-Normal [†]	$\frac{1}{x \beta \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{\log x - \alpha}{\beta}\right)^2}$	[–9, 1]	[–2, 3]
Scaled Beta [‡]	$\frac{x^{\alpha-1} (1 - x^\alpha)^{\beta-1}}{B(\alpha, \beta)}$	[–3, 3]	[0, 3]
Scaled Kumaraswamy [‡]	$\alpha \beta x^{\alpha-1} (1 - x^\alpha)^{\beta-1}$	[–3, 3]	[0, 3]

^{*} Priors for α, β in the applied model were defined as uniform on the log-scale; e.g. [–1, 3] corresponds to $[e^{-1}, e^3] \approx [0.37, 20]$.

[†] Distributions spanning $[0, \infty)$ were truncated to $[0, 50]$ years. [‡] Distributions spanning $[0, 1]$ were scaled to span $[0, 50]$ years.

Likelihoods functions Under the model for $p(z|s)$, we define the likelihoods for empirical estimates of the mean $\hat{\mathbb{E}}_i[z|s]$ and proportions $\hat{\mathbb{P}}_i[a < z < b|s]$ from a given survey with sample size N_i as, respectively: the Gamma-equivalent of Normal sampling error for the mean (since $\hat{\mathbb{E}} > 0$), and the Beta-approximation of the Binomial distribution (for differentiability):

$$\hat{\mathbb{E}}_i[z|s; N_i] \sim \text{Gamma}(\alpha = N_i / \mathbb{CV}_i^2[z|s], \beta = N_i / \mathbb{CV}_i^2[z|s] / \mathbb{E}[z|s]) \quad (\text{A.1})$$

$$\hat{\mathbb{P}}_i[a < z < b|s; N_i] \sim \text{Beta}(\alpha = N_i \mathbb{P}_i[a < z < b|s], \beta = N_i (1 - \mathbb{P}_i[a < z < b|s])) \quad (\text{A.2})$$

To validate these likelihood functions, we simulated 10,000 stochastic samples of z with sample sizes $N_i = \{30, 100, 300, 1000\}$ given known $p(z|s)$ under $\mathbb{E}[x] = 5$, $\mathbb{CV}[x] = \{0.5, 1, 1.5\}$, and all 6 distribution families. We then compared the distribution of stochastic estimates of $\hat{\mathbb{E}}[z|s; N_i]$ and $\hat{\mathbb{P}}[a < z < b|s; N_i]$ from each sample with the analytic likelihood functions: Eqs. (A.1) and (A.2). For proportions, we considered $\mathbb{P}[0 < z < 1]$, $\mathbb{P}[1 < z < 5]$, and $\mathbb{P}[5 < z < 9]$, arbitrarily.

Figures A.1–A.4 illustrate these comparisons, where we observe good agreement between the stochastic and analytic distributions across all conditions. Note that stochastic distributions for $\hat{\mathbb{P}}[a < z < b|s; N_i]$ are discrete (more visible at low N_i) because there are a finite number simulated respondents whose durations can fall in the specified interval $[a, b]$.

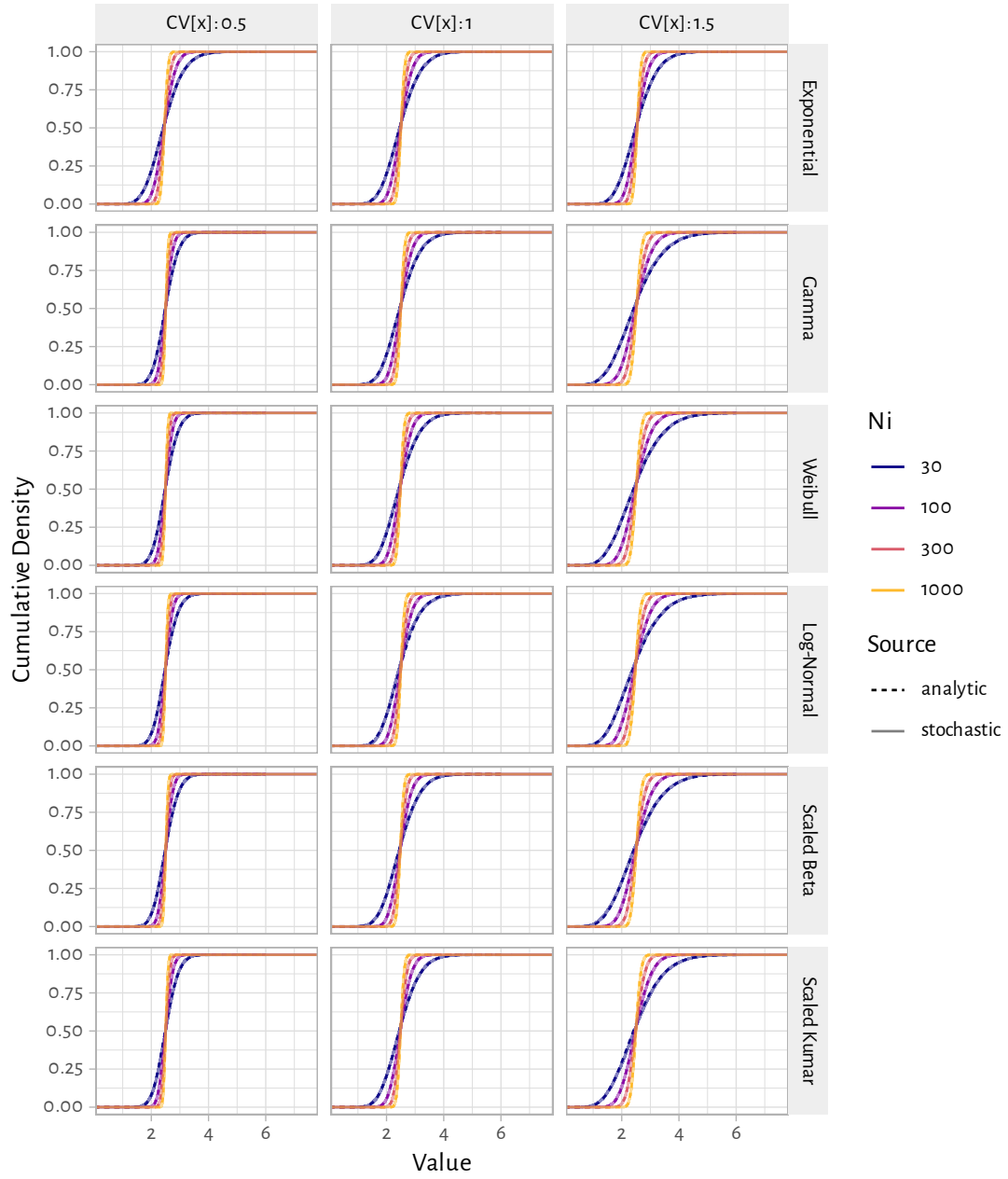


Figure A.1: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{E}[\mathbf{z}|\mathbf{s}]$, for different simulated survey sizes N_i and latent distribution families for $p(x)$

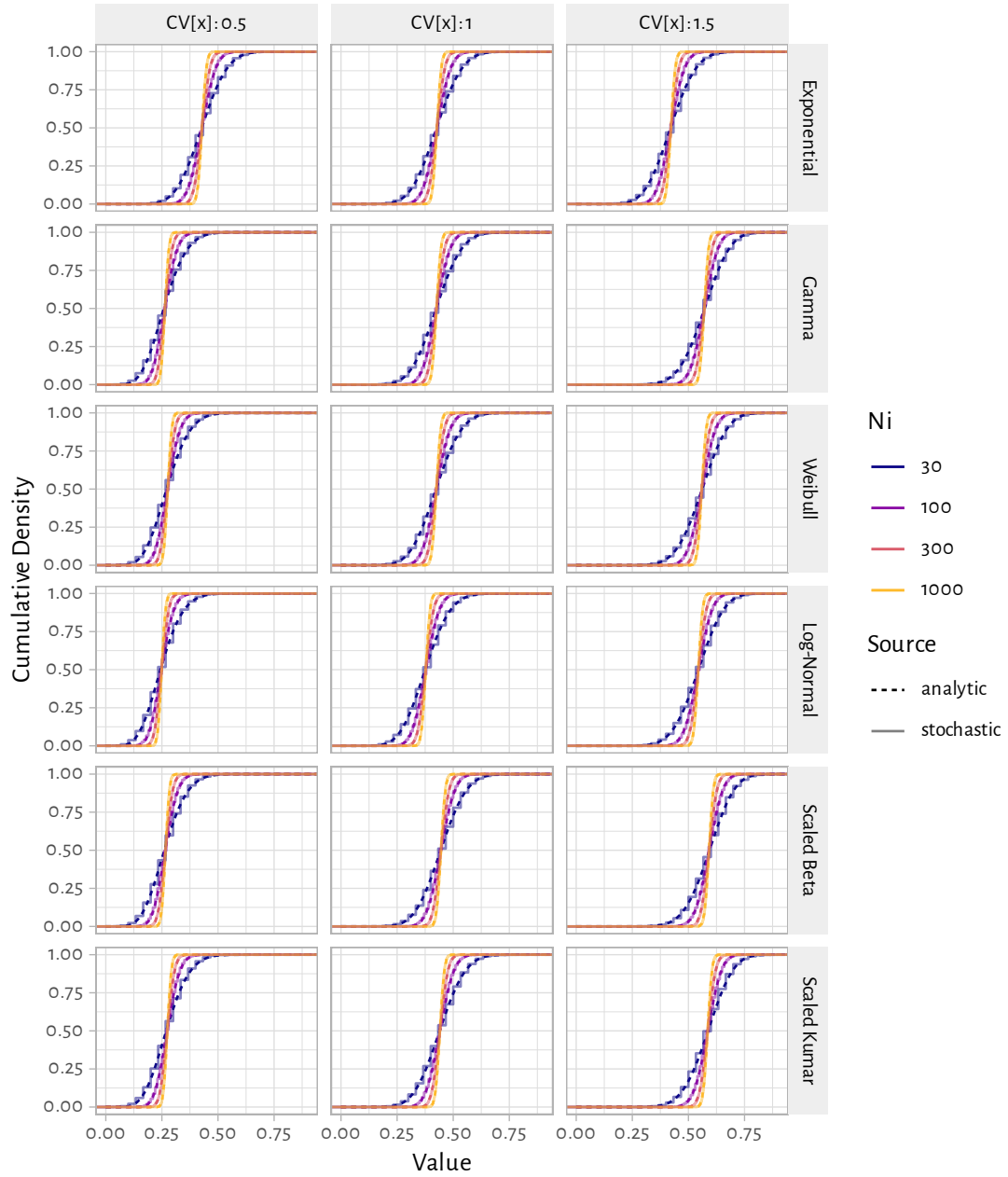


Figure A.2: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{P}[\mathbf{0} < \mathbf{z} < \mathbf{1} | \mathbf{s}]$, for different simulated survey sizes N_i and latent distribution families for $p(x)$

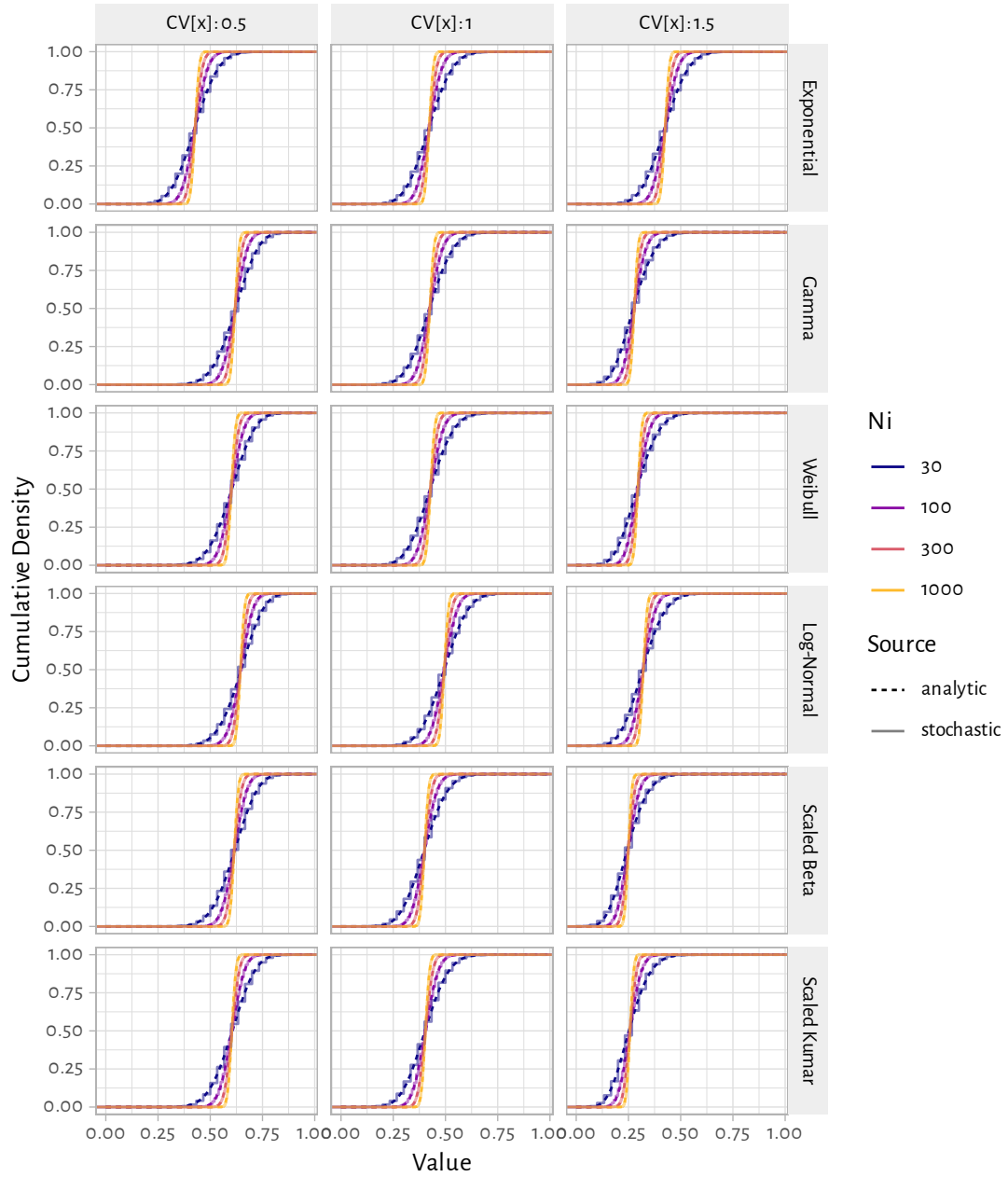


Figure A.3: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{P}[1 < z < 5|s]$, for different simulated survey sizes N_i and latent distribution families for $p(x)$

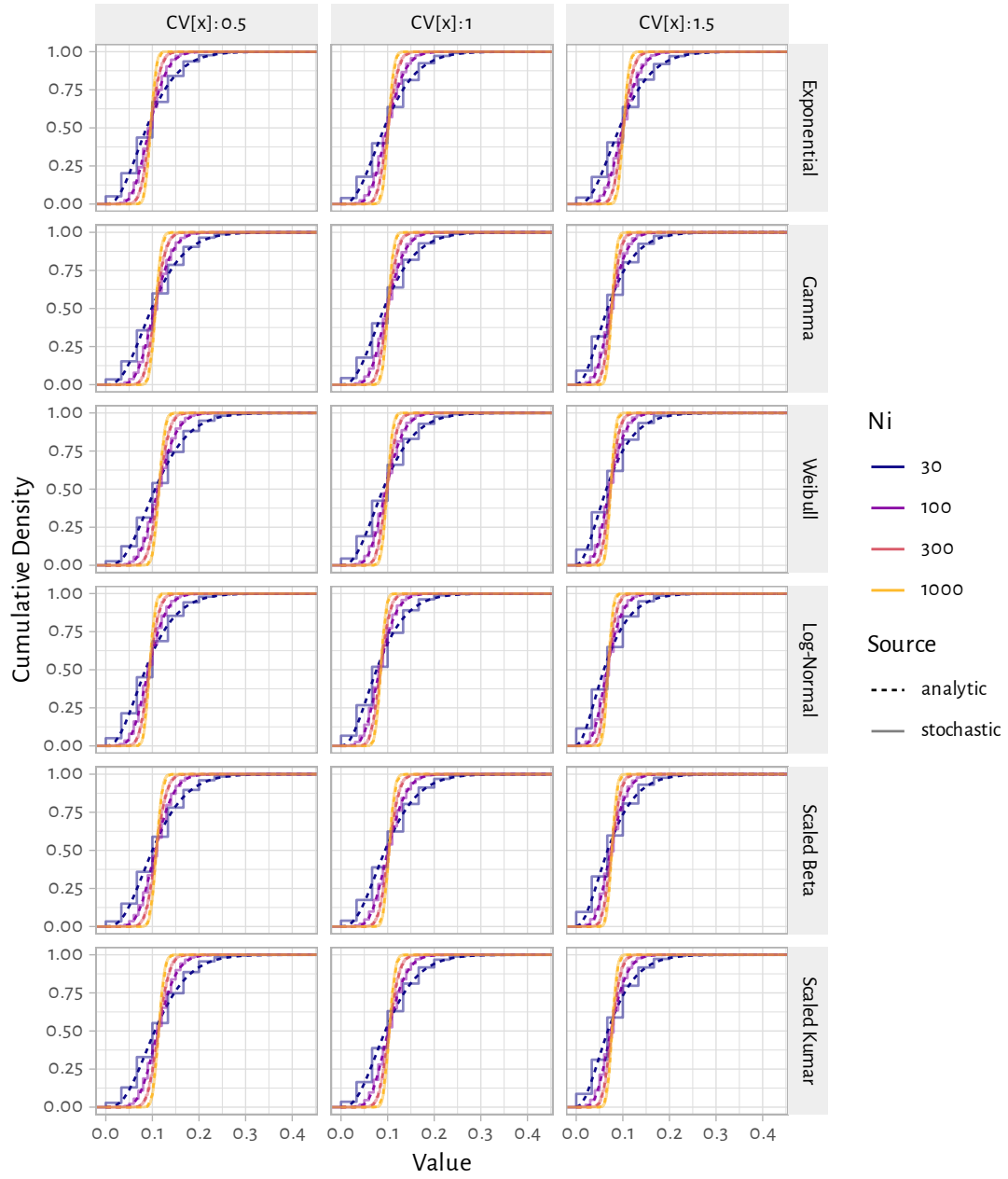


Figure A.4: Comparison of analytic likelihood functions and distribution of 10,000 stochastic estimates of $\mathbb{P}[5 < z < 9|s]$, for different simulated survey sizes N_i and latent distribution families for $p(x)$

A.2 Illustrative analytic results

Figure A.5 repeats Figure 2 for 5 distribution families considered (Table A.1) — i.e. showing distributions of total (x) and censored (z) durations among the source, active, and sampled populations for each family, all having $\mathbb{E}[x] = 5$ and $\text{CV}[x] = \{0.5, 1, 1.5\}$. We omit the Exponential family as it is a special case of both Gamma and Weibull with $\text{CV}[x] = 1$. Figure A.6 further illustrates performance of different $\mathbb{E}[x]$ estimators considered in [1] across a range of $\text{CV}[x]$ and the same 5 distribution families.

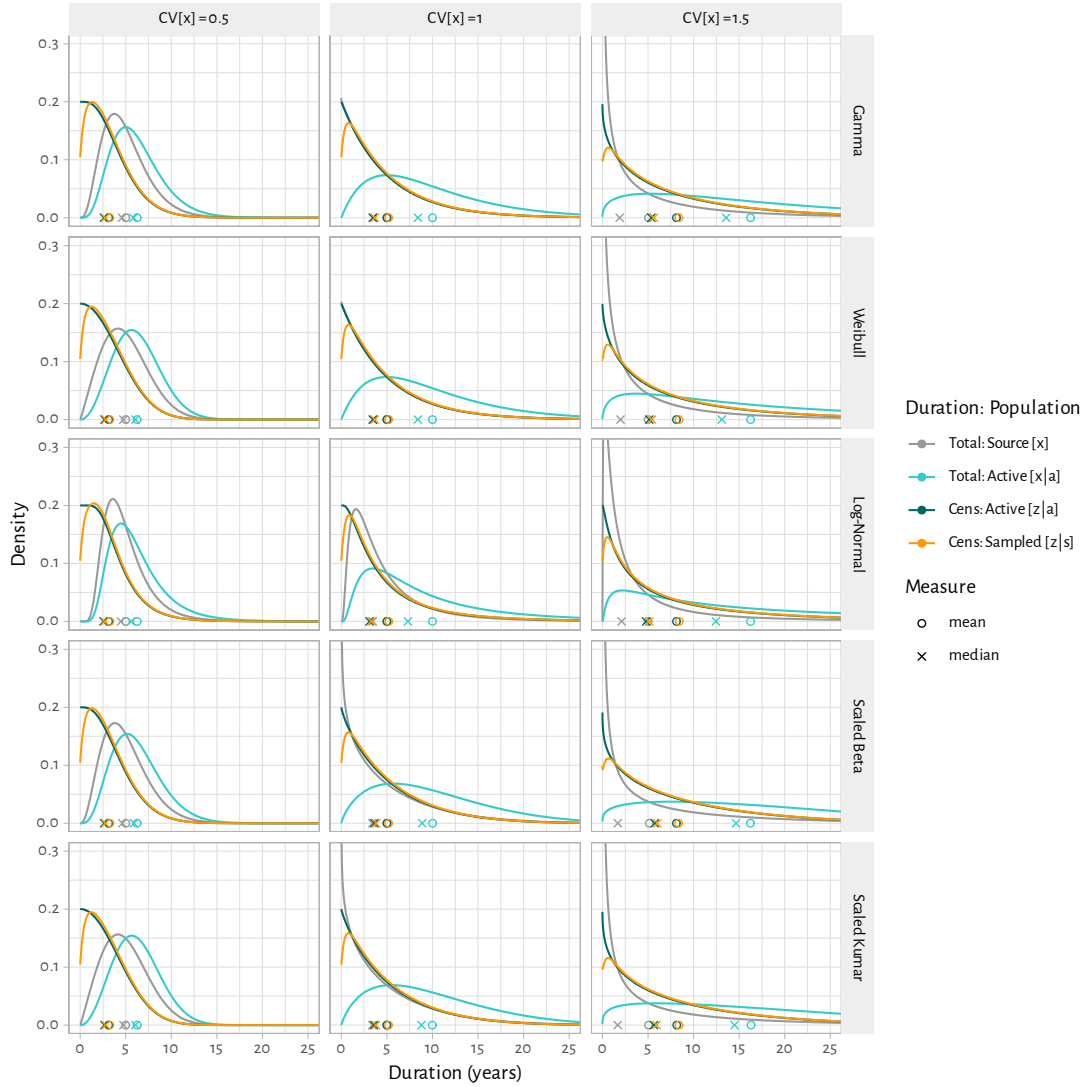


Figure A.5: Illustrative distributions of total and censored durations among the source, active, and sampled populations for different x distribution families (columns) and 3 coefficients of variation $\text{CV}[x]$ (rows)

Note: true $\mathbb{E}[x] = 5$; probability of being sampled modelled per Eq. (7) with $\rho_s = 0.5$, $\tau_{95} = 1$.

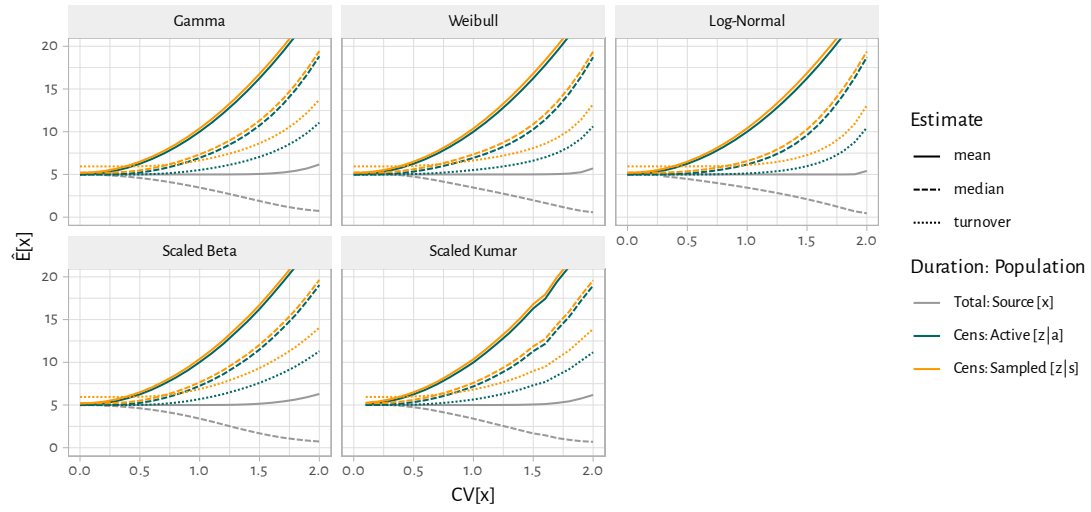


Figure A.6: Performance of $E[x]$ estimators considered in [1] across a range of $CV[x]$ and distribution families

Note: true $E[x] = 5$; probability of being sampled modelled per Eq. (7) with $\rho_s = 0.5$, $\tau_{95} = 1$; means and medians for censored data (z) are doubled per [1].

A.3 Application

A.3.1 Aggregate data from a single survey

Table A.2 summarizes the RDS-adjusted sex work duration data reported in [2], while Figure A.7 illustrates the posterior distributions inferred by fitting our Bayesian model to those data.

Table A.2: RDS-adjusted proportions of durations selling sex among Swati female sex workers in [2] (N = 313)

Duration [†] (years)	Proportion (%)			N_i Eff. [‡]
	Crude	RDS	(95% CI)	
0–3	28.8	38.3	(27.5, 49.1)	77
3–6	31.3	32.1	(23.6, 40.7)	113
6–11	26.2	20.2	(13.2, 27.1)	129
11+	13.7	9.4	(4.4, 14.4)	135

[†] Reported strata in [2] were: "under 2, 3–5, 6–10, 11 or more", which we corrected here to avoid implied gaps between strata, [‡] Effective sample size per strata calculated to match RDS-adjusted 95% CI.

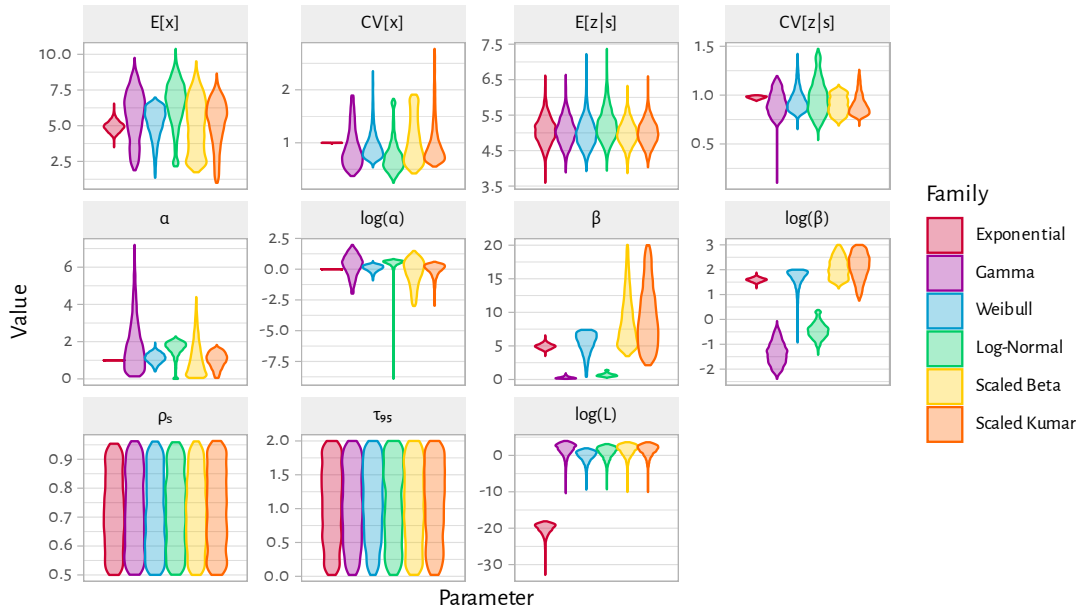


Figure A.7: Posterior distributions of model parameters and summary statistics when fitting to the RDS-adjusted Eswatini data (Table A.2)

A.3.2 Aggregate data from a global review

Data The manually extracted data from Table S2 in [1] are provided as supplementary file `Fazit02012.csv`. We extracted all reported duration means $\mathbb{E}[z|s]$ and interval proportions $\hat{\mathbb{P}}[a < z < b|s]$ (including medians and maximum durations), but dropped from our analyses for simplicity: standard deviations, minimum durations, and durations reported as "greater/less than X". We also dropped estimates from the following 9 sources due to the female sex worker (FSW) population definition (it's not clear whether these studies were included in the quantitative syntheses of [1]):

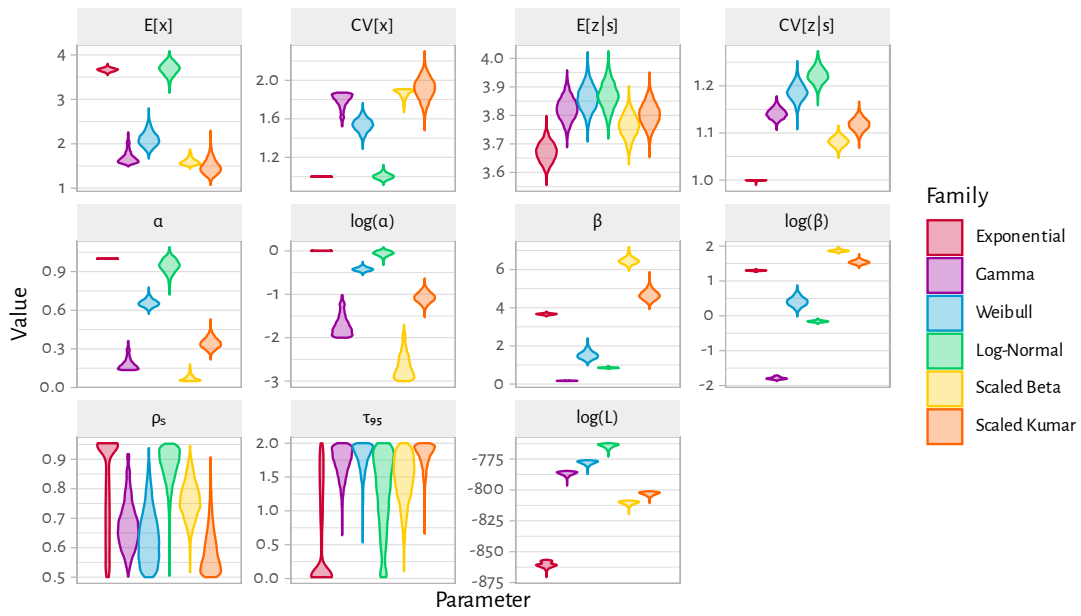
- Anzala (1995) Kenya: FSW in a cohort who acquired HIV
- Ward (2006) UK: FSW who retired and were followed-up
- Pineda (1995) Spain: FSW working for at least 21 months
- Strathdee (2008) Mexico: duplicate of Loza (2010) Mexico
- MOH (2003) Cambodia: not clearly FSW (e.g. karaoke workers)
- MOH (2007) Cambodia: not clearly FSW (e.g. karaoke workers)
- Gu (2009) China: FSW who inject drugs
- Silverman (2006) India: women trafficked into selling sex
- Simoes (1993) India: women incarcerated for selling sex

Before synthesizing the extracted data via our Bayesian model, we also adjusted the effective sample sizes for each study N_i to reduce the influence of very large studies via:

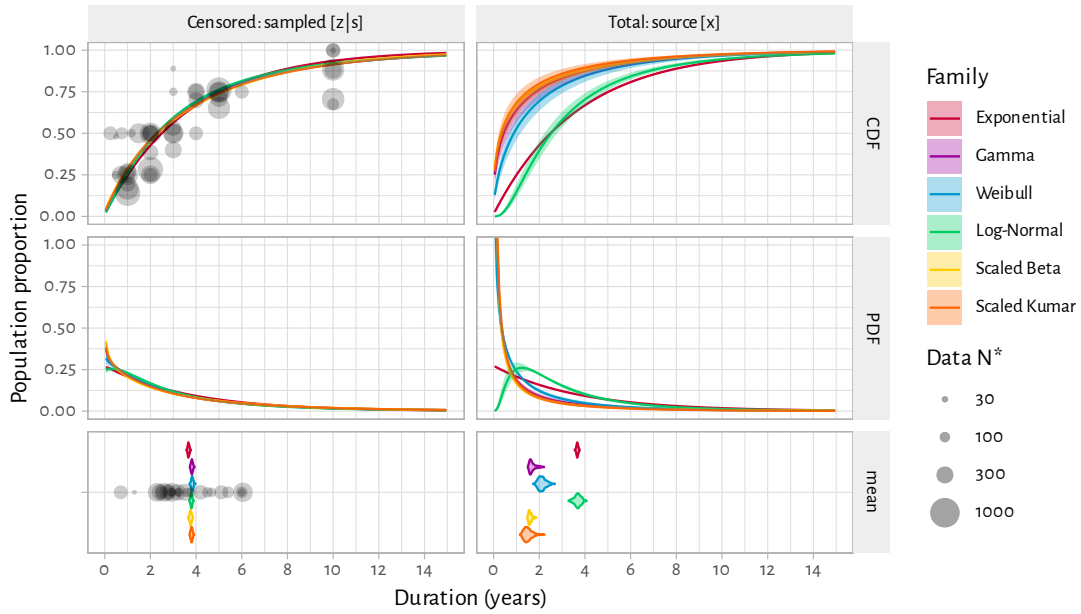
$$N_i^* = \frac{1}{\frac{1}{N_0} + \frac{1}{N_i}}, \quad N_0 = 1000 \quad (\text{A.3})$$

This adjustment loosely approximates a random effects approach by weighting each study more equally.

Results Figures A.8–A.11 illustrate the posterior distributions of model parameters, summary statistics, and cumulative/density duration distributions and means versus the data for female sex workers reviewed in [1].



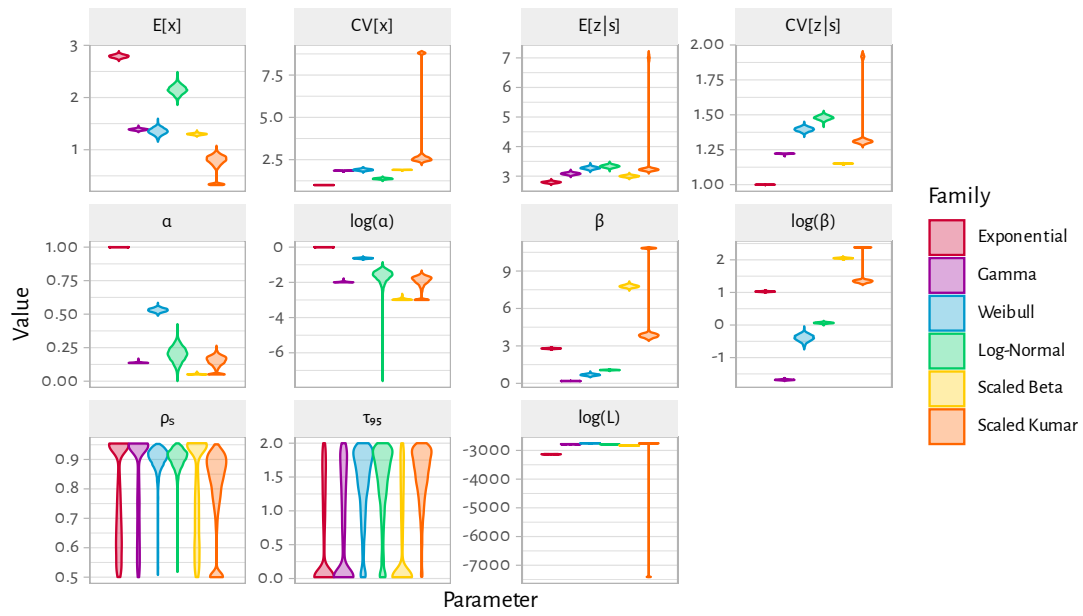
(a) Model parameter and summary statistics



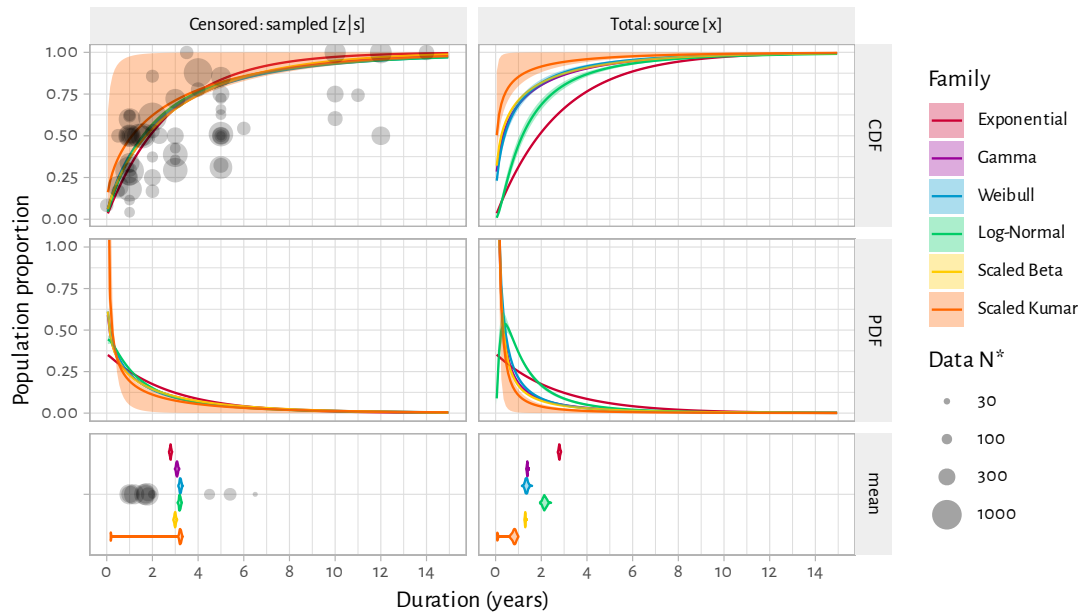
(b) Modelled cumulative (CDF) and density (PDF) distributions and means versus empirical data

Figure A.8: Model posteriors when fitting to empirical aggregate female sex worker duration data reviewed in [1] for **Africa**

Coloured lines & shaded ribbons denote mean & 95% CI distribution values; gray circles denote aggregate empirical estimates



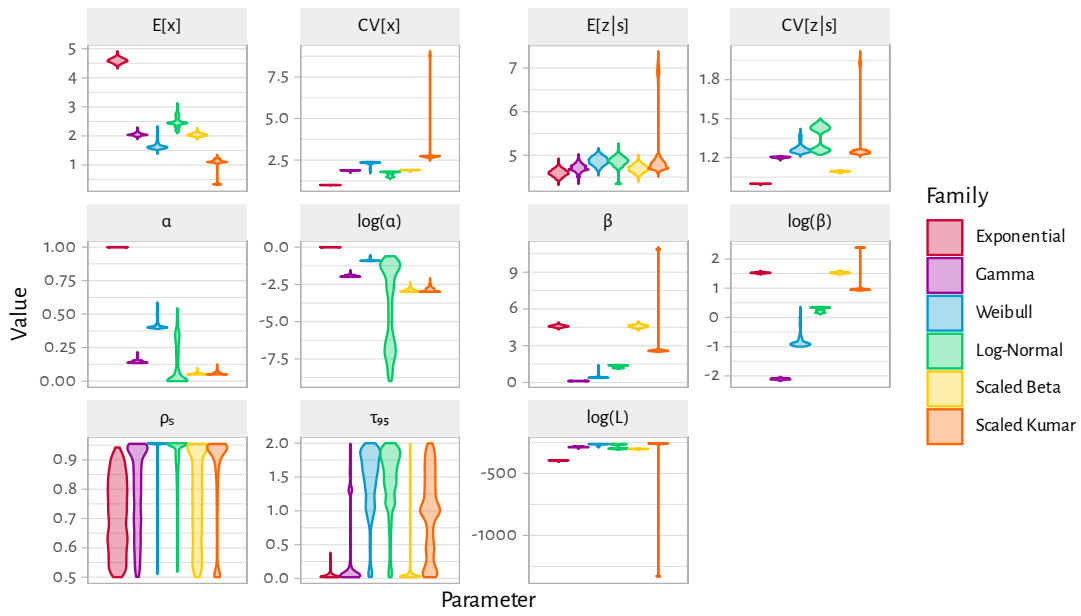
(a) Model parameter and summary statistics



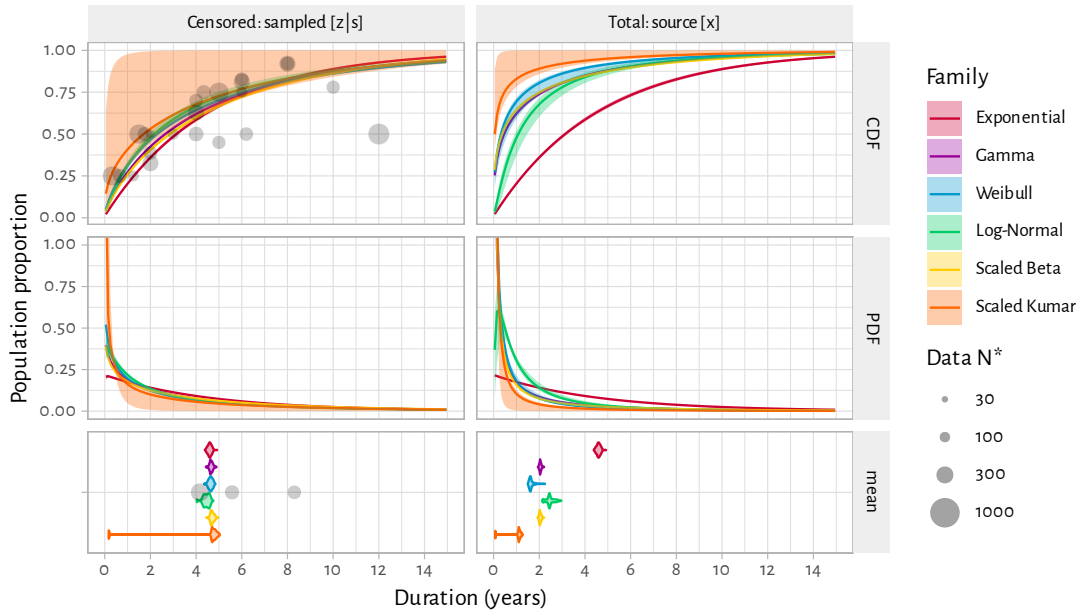
(b) Modelled cumulative (CDF) and density (PDF) distributions and means versus empirical data

Figure A.9: Model posteriors when fitting to empirical aggregate female sex worker duration data reviewed in [1] for **Asia**

Coloured lines & shaded ribbons denote mean & 95% CI distribution values; gray circles denote aggregate empirical estimates



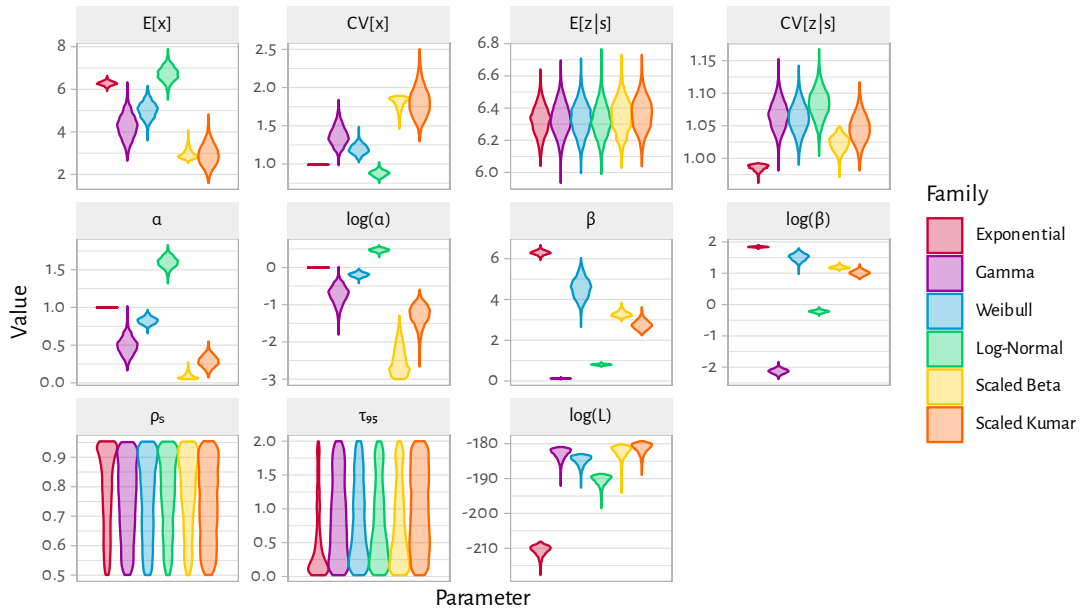
(a) Model parameter and summary statistics



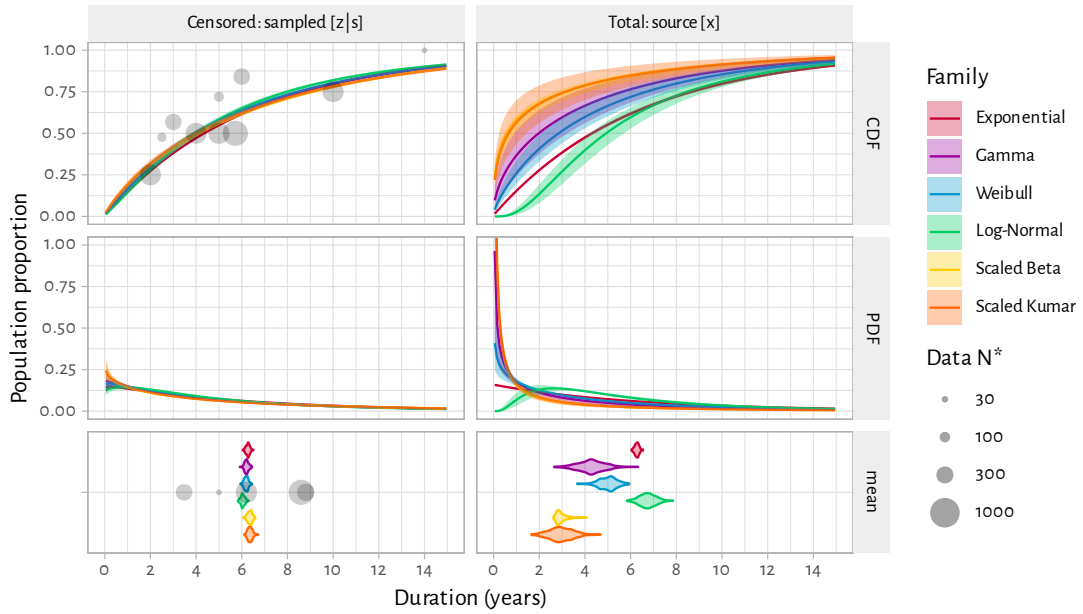
(b) Modelled cumulative (CDF) and density (PDF) distributions and means versus empirical data

Figure A.10: Model posteriors when fitting to empirical aggregate female sex worker duration data reviewed in [1] for **Europe**

Coloured lines & shaded ribbons denote mean & 95% CI distribution values; gray circles denote aggregate empirical estimates



(a) Model parameter and summary statistics



(b) Modelled cumulative (CDF) and density (PDF) distributions and means versus empirical data

Figure A.11: Model posteriors when fitting to empirical aggregate female sex worker duration data reviewed in [1] for **Latin America**

Coloured lines & shaded ribbons denote mean & 95% CI distribution values; gray circles denote aggregate empirical estimates